LAP2 — LAP2 TASK 1: MOBILE APP DEVELOPMENT

MOBILE APPLICATION DEVELOPMENT USING C# — C971

PRFA — LAP2

COMPETENCIES

4026.2.1 : Introduction to Mobile Development
The graduate develops a simple mobile application using an integrated development environment (IDE).

4026.2.2 : User Interfaces
The graduate develops a user interface for a mobile application.

4026.2.3 : Managing Data
The graduate creates solutions to store, retrieve, and transmit mobile application data.

4026.2.4 : Supporting Devices and Platforms
The graduate develops an application that accounts for different platforms and device conditions.

INTRODUCTION

As a competent mobile application developer, your understanding of mobile application structure and design will help you to develop applications to meet customer requirements.

For this task, you will develop a multiple-screen mobile application for WGU students to track their academic terms, courses associated with each term, and assessments associated with each course. The application will allow students to enter, edit, and delete term, course, and assessment data. It should provide summary and detailed views of courses for each term, provide notifications for upcoming performance and objective assessments, and notify students of deadlines and tracking milestones even after the application is closed. You will create a wireframe as a visual guide, representing the skeletal framework of the application. Your application will use an SQLite database using the SQLIte ORM located at SQLite-net.

You will share this project to your GitLab repository and backup regularly. Use the GitLab link in the web links section to create your Gitlab project in the WGU GitLab space, and prior to starting your work, reference the "GitLab How-To" web link to set up your project.

This task will allow you to demonstrate your ability to apply the skills learned in the course and help you to apply these skills in a familiar, real-world scenario.

The assessment must be submitted using the NET Multi-platform App UI (MAUI) framework. For this assessment, you are welcome to use these plugins or plugins referenced at these sites:

.NET Multi-platform App UI (.NET MAUI) Community Toolkit Documentation
DatePicker

.NET MAUI Local Databases
Picker
Plugin.LocalNotification
Share
SQL tite-net

Any controls specified as part of the Microsoft .NET MAUI Toolkit are allowed in the project. All other external plugins and libraries are not allowed.

REQUIREMENTS

Your submission must represent your original work and understanding of the course material. Most performance assessment submissions are automatically scanned through the WGU similarity checker. Students are strongly encouraged to wait for the similarity report to generate after uploading their work and then review it to ensure Academic Authenticity guidelines are met before submitting the file for evaluation. See Understanding Similarity Reports for more information.

Grammarly Note:
Professional Communication will be automatically assessed through Grammarly for Education in most performance assessments before a student submits work for evaluation. Students are strongly encouraged to review the Grammarly for Education feedback prior to submitting work for evaluation, as the overall submission will not pass without this aspect passing. See Use Grammarly for Education Effectively for more information.

Microsoft Files Note:
Write your paper in Microsoft Word (.doc or .docx) unless another Microsoft product, or pdf, is specified in the task directions. Tasks may not be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc. All supporting documentation, such as screenshots and proof of experience, should be collected in a pdf file and submitted separately from the main file. For more information, please see Computer System and Technology Requirements.

You must use the rubric to direct the creation of your submission because it provides detailed criteria that will be used to evaluate your work. Each requirement below may be evaluated by more than one rubric aspect. The rubric aspect titles may contain hyperlinks to relevant portions of the course.

Note: You will be submitting pictures, graphics, and/or diagrams. Each picture, graphic, and/or diagram should be an attachment no larger than 30 MB in size. Diagrams should be original and may be hand-drawn or drawn using a graphics program. CAD programs should not be used because attachments will be too large.

A. Create a GitLab repository for your subgroup and project. Commit with a message and push when you complete *each* requirement listed in parts B1, B2, B3, C1, C2, C3, C4, C5, and C6. Submit **both** of the following:
   - the GitLab repository URL
   - a copy of the repository branch history retrieved from your repository, which must include *all* the commit messages and dates

   Note: You may commit and push whenever you want to back up your changes, even if a requirement is not yet complete. Wait until you have completed all the following prompts before you create your copy of the

repository branch history.

B. Draw a low-fidelity wireframe for your mobile application that includes all of the following requirements:
   1. Add as many academic terms as needed; *each* term should include the addition, editing, deletion, and storage of all the following:
      - the term title
      - start and anticipated end dates (using a date picker)
   2. Add **six** courses for *each* term. For *each* course, include the addition, editing, deletion, and storage of *all* the following course information:
      - course name
      - start and anticipated end dates (using a date picker)
      - course status (using a picker)
      - course instructor's information: name, phone, and email
      - add, share, and display optional notes
      - set notifications for the start and end dates of each course
      - display of a detailed view of each course, including the due date
      - editing of the detailed course view screen
   3. Add **two** assessments for *each* course. *Each* assessment should include the addition, editing, deletion, and storage of *all* the following assessment information:
      - **one** objective assessment
      - **one** performance assessment
      - names of **both** assessments
      - notifications for the start and end dates of **both** assessments

   *Note: For any editing dates shown in the wireframe, use a date picker and make it clear the dates being edited are using a date picker and **not** a text box for data editing. Additionally, when using other pickers in editing mode, be sure those are clearly pickers and **not** textboxes.*

C. Create, within the .NET Multi-platform App UI (MAUI) framework, a mobile application aligned to the wireframe drawn in part A. Include all of the following features:
   1. Provide an interface for *all* the following information for as many academic terms as needed:
      - academic term title (e.g., Term 1, Term 2, Spring term)
      - start and end dates (using a date picker)

      Note: You are welcome to use the following plugin for this step: DatePicker

   2. Provide an interface that allows the user to access *all* the following features for *each* academic term:
      - add and display a list of **six** courses for *each* term
      - display a detailed view of each term, including *all* the information from part C1
   3. Provide an interface that allows the user to access and edit *all* the following details for *each* course:
      - course title
      - start and anticipated end dates (using a date picker)

      Note: You are welcome to use the following plugin for this step: DatePicker

      - course status (e.g., in progress, completed, dropped, plan to take) (using a picker)

      Note: You are welcome to use the following plugin for this step: Picker

- the course instructor's name, phone number, and email address; include validation to prevent the user from saving a null value (e.g., an invalid email address

4. Create features that allow the user to do *all* of the following for *each* course:
   - enter **two** assessments: **one** performance assessment and **one** objective assessment
   - add and display optional notes
   - enter, edit, and delete course information
   - display an editable detailed view of the course, including the due date
   - set alerts (e.g., notifications) for the start and end date of the course

   Note: You are welcome to use the following plugin for this step: Plugin.LocalNotification

   - share notes via a sharing feature (e.g., email, SMS)

   Note: You are welcome to use the following plugins for this step: Share

5. Provide an interface for the user to do *all* of the following for *each* assessment:
   - include the names and due dates
   - enter, edit, and delete assessment information
   - set notifications for anticipated start and end dates of each assessment

   Note: You are welcome to use the following plugin for this step: Plugin.LocalNotification

6. Write code to create a set of data for evaluation purposes, including the following details:
   - **one** term and **one** course from part C3
   - the **two** assessments from part C4 for that course
   - the following name, phone number, and email address as the course instructor contact information:
   - Anika Patel
   - 555-123-4567
   - anika.patel@strimeuniversity.edu

D. Examine the wireframe from part B to determine any changes that you made during the development of the mobile application.
   1. Explain the reasons for *any* changes made during the development of the mobile application by adding comments within the wireframe. If no changes were made, include that comment.
   2. Confirm that the wireframe aligns with the completed mobile application from part C *after* the inclusion of the comments from part D1.

E. Provide a complete and operational source file for your mobile application.

F. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ( )
File size limit: 200 MB
File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A:GITLAB REPOSITORY

NOT EVIDENT

A GitLab repository is not provided.

APPROACHING COMPETENCE

The URL for the GitLab repository is missing or incorrect, or the copy of the repository branch history does not include *all* the commit messages and dates.

COMPETENT

An accurate URL for the GitLab repository is provided and the copy of the repository branch history includes *all* the commit messages and dates.

B1:ACADEMIC TERMS IN WIREFRAME

NOT EVIDENT

The low-fidelity wireframe does not allow for the inclusion of academic terms.

APPROACHING COMPETENCE

The low-fidelity wireframe includes as many academic terms as needed, but it does not allow for the addition, editing, deletion, and storage of term titles or the start and anticipated end dates associated with each term. Or dates are not entered using a date picker, or it is not verified that all start dates for terms are set before the anticipated end dates.

COMPETENT

The low-fidelity wireframe includes as many academic terms as needed, the addition, editing, deletion, and storage of term titles, and the start and anticipated end dates associated with each term. Dates are entered using a date picker and it is verified that all start dates for terms are set before the anticipated end dates.

B2:COURSES IN WIREFRAME

NOT EVIDENT

The low-fidelity wireframe does not allow for the inclusion of *any* courses for *each* term.

APPROACHING COMPETENCE

The low-fidelity wireframe allows for the inclusion of only 1–5 courses for *each* term, or it does not include the information associated with *each* course. Or the wireframe does not allow for the addition, editing, deletion, or storage of *all* additional given course information. Or it is not verified that start dates for *each*

COMPETENT

The low-fidelity wireframe allows for the inclusion of 6 courses for *each* term and the information associated with *each* course. The wireframe for *each* course allows for the addition, editing, deletion, and storage of *all* additional given course information. It is verified that start dates for *each* course are set before the anticipated end dates.

course are set before the anticipated end dates.

B3:ASSESSMENTS IN WIREFRAME

NOT EVIDENT

The low-fidelity wireframe does not allow for the inclusion of any assessments for *each* course.

APPROACHING COMPETENCE

The low-fidelity wireframe allows for the inclusion of only 1 assessment for *each* course, or it does not include 1 objective assessment and 1 performance assessment for each course. Or it does not include the addition, editing, deletion, or storage of *all* the given assessment information.

COMPETENT

The low-fidelity wireframe allows for the inclusion of 2 assessments for *each* course, 1 of which is an objective assessment and 1 of which is a performance assessment, and includes the addition, editing, deletion, and storage of *all* the given assessment information.

C1:INTERFACE FOR TERMS

NOT EVIDENT

The mobile application interface does not allow the user to enter any information for any academic terms.

APPROACHING COMPETENCE

The mobile application interface does not allow the user to enter the title of each academic term or all start and end dates for each term. Or dates are not entered using a date picker, or it is not verified that all start dates for terms are set before the anticipated end dates.

COMPETENT

The mobile application interface allows the user to enter the title of each academic term and all start and end dates for each term. Dates are to be entered using a date picker, and it is verified that all start dates for terms are set before the anticipated end dates.

C2:TERM INTERFACE FOR USER

NOT EVIDENT

The mobile application does not provide an interface that allows the user access to any features for the courses.

APPROACHING COMPETENCE

The mobile application interface does not allow the user to add or display all 6 courses for *each* term, or it displays only a limited view of the data from part C1.

COMPETENT

The mobile application interface allows the user to add and display 6 courses for *each* term and to display a detailed view of each term that includes *all* the information from part C1.

C3:COURSE INTERFACE

NOT EVIDENT

The mobile application interface does not allow the user to access or edit *any* of the given details for *each* course.

APPROACHING COMPETENCE

The mobile application interface allows the user to access, but not to edit, the given details for *each* course. Or the application does not use a date picker for all start and anticipated end dates, or does not use a picker to enter the course status. Or it is not verified that all start dates for *each* course are set before the anticipated end dates. Or the interface does not include validation to prevent the user from saving a null value.

COMPETENT

The mobile application interface allows the user to access and edit *all* of the given details for *each* course. The application uses a date picker for all anticipated start and end dates and uses a picker to enter the course status. It is verified that all start dates for each course are set before the anticipated end dates. The interface includes validation to prevent the user from saving a null value.

C4:COURSE FEATURES FOR USER

NOT EVIDENT

The features do not allow the user to do *any* of the given activities for any of the courses.

APPROACHING COMPETENCE

The features allow the user to enter only 1 assessment for each course, or it allows the user to enter 2 assessments for each course but with no distinction between the objective assessment and the performance assessment. Or the created features do not allow the user to do *all* of the given activities.

COMPETENT

The features allow the user to enter 2 assessments, 1 of which is a performance assessment and 1 of which is an objective assessment, for each course and allow the user to do *all* of the given activities.

C5:ASSESSMENT INTERFACE

NOT EVIDENT

The assessment interface does not allow the user to do *any* of the given activities.

APPROACHING COMPETENCE

The assessment interface does not allow the user to do *all* of the given activities.

COMPETENT

The assessment interface allows the user to do *all* of the given activities.

C6:DATA SET

NOT EVIDENT

The code to create a set of data is not provided.

APPROACHING COMPETENCE

The code creates a set of data for evaluation purposes, but it does not include either 1 term or 1 course from part C3, or 1 or more assessments from part C4. Or all the course instructor's given contact information is not accurately provided.

COMPETENT

The code creates a set of data for evaluation purposes, including 1 term and 1 course from part C3 and the 2 assessments from part C4. The course instructor's given contact information is accurately provided.

D1:CHANGES TO WIREFRAME

NOT EVIDENT

The wireframe from part A contains no comments about the changes made during the development of the mobile application.

APPROACHING COMPETENCE

The wireframe from part A includes comments that identify changes made during the development but do not explain the reasons for the changes.

COMPETENT

The wireframe from part A includes comments that explain the reasons for *any* changes made during the development of the mobile application. Or there is a comment stating no changes were made.

D2:CONFIRM WIREFRAME MATCHES

NOT EVIDENT

The submission does not confirm the alignment of the wireframe and the mobile application.

APPROACHING COMPETENCE

The wireframe aligns with the completed mobile application from part C, but only *before* the inclusion of the comments from part D1.

COMPETENT

The wireframe aligns with the completed mobile application from part C, which includes the comments from part D1.

E:SOURCE FILE

NOT EVIDENT

A source file for the mobile application is not provided.

APPROACHING COMPETENCE

The source file for the mobile application is incomplete or cannot run on any platform as a .NET MAUI application.

COMPETENT

The source file for the mobile application is complete and runs on any platform as a .NET MAUI application.

F:PROFESSIONAL COMMUNICATION

NOT EVIDENT

This submission includes pervasive errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation, negatively impacting the professional quality and clarity of the writing. Specific errors have been identified by Grammarly for Education under the Correctness category.

APPROACHING COMPETENCE

This submission includes substantial errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation. Specific errors have been identified by Grammarly for Education under the Correctness category.

COMPETENT

This submission includes satisfactory use of grammar, sentence fluency, contextual spelling, and punctuation, which promote accurate interpretation and understanding.

WEB LINKS

.NET Multi-platform App UI (.NET MAUI) Community Toolkit Documentation

DatePicker

WGU GitLab Environment Login

.NET MAUI Local Databases

Picker

Plugin.LocalNotification

Share

SQLite-net

SUPPORTING DOCUMENTS

C971 WGU GitLab Environment.docx

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